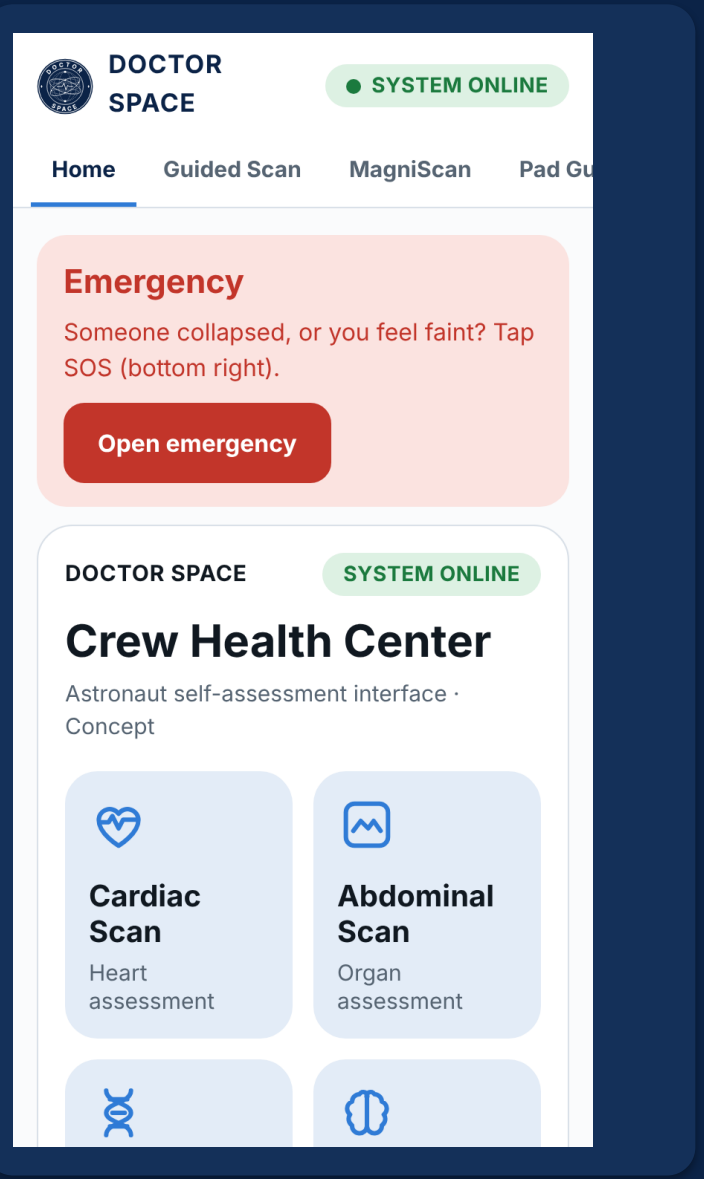
DOCTOR SPACE

Crew Health Center

A magnetic-pad ultrasound guide that lets a crew member who is not a doctor check the heart and belly, read the result, and know the next step.

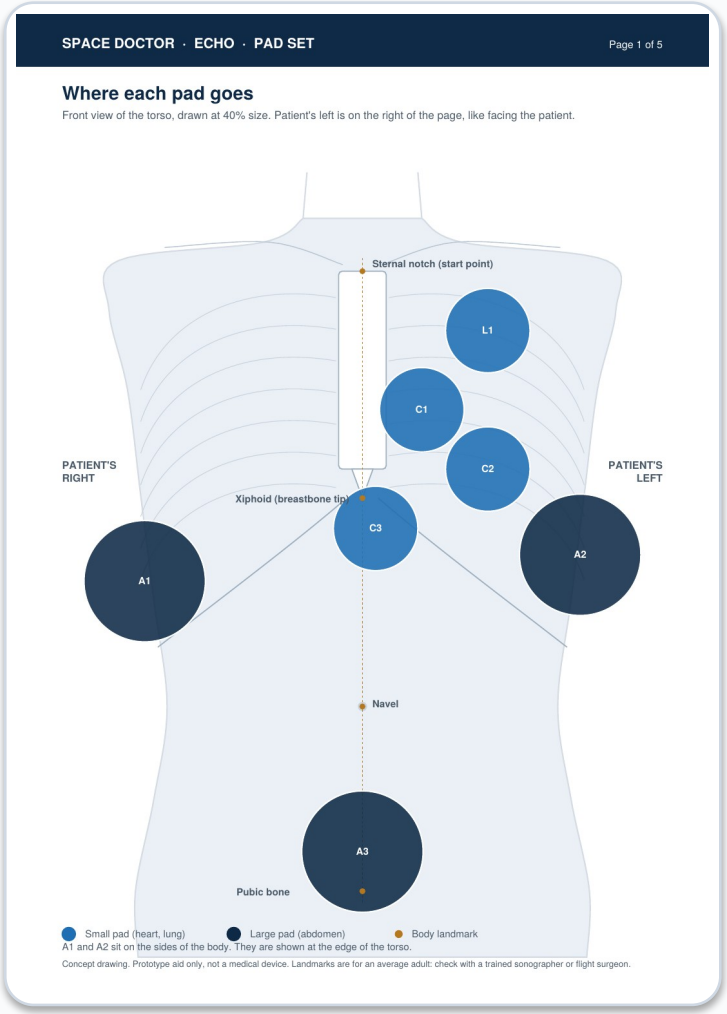
Team: DOCTOR SPACE
Member: Alfathitamara Oktalia Sasti
Challenge: DOCTOR SPACE
NASA Space Apps Challenge 2026

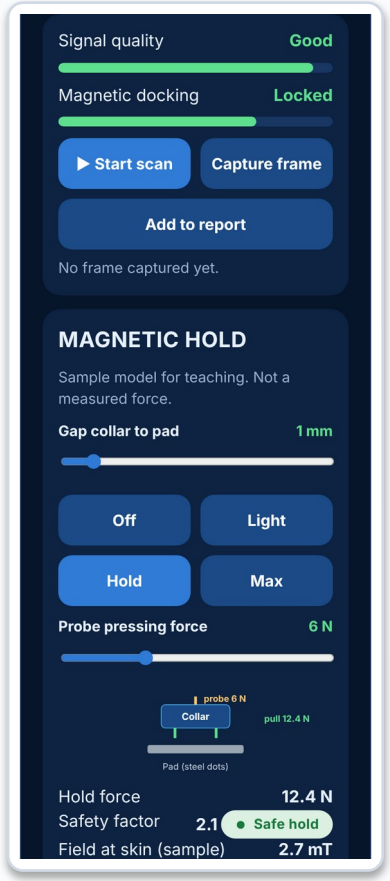


Wearable magnetic ultrasound

Magnets hold the probe in the right place. The crew member does not have to find it by feel.

- **The gap.** On a long mission a doctor may be days from Earth, and a doctor may not be on board at all.
- **Sticker pads.** A printed sheet of anatomical pads goes on the skin. Each pad has a window for the probe and a few steel dots.
- **Magnetic collar.** Small magnets in a collar on the probe dock to the steel dots. They hold position and angle, not pressure.
- **Same spot every time.** Pads stay on, so today's image can be compared with the last one.
- **Low cost to try.** Printed sheet, steel dots, small magnets. A first prototype fits on a table.



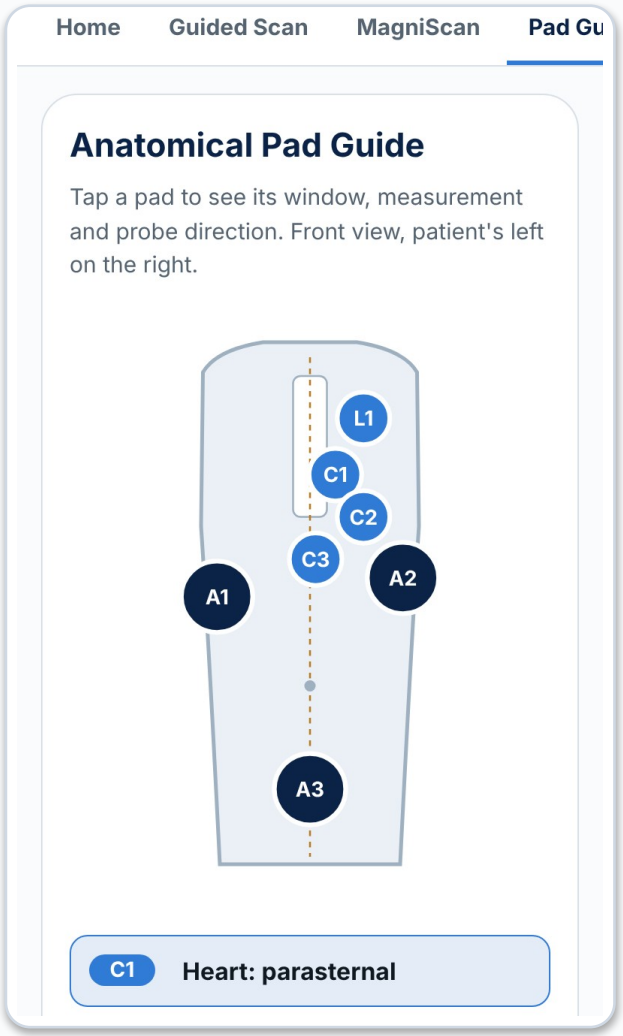


MagniScan hold model: gap, force and field at skin, with a safety factor shown.

Concept prototype. Simulated data and illustrations. Not a validated medical device.

Where the probe goes

Seven pads cover the heart, lung and belly. Each one has a landmark that anyone can find.



C1 / C2 / C3	Heart: beside the breastbone, under the left nipple line, and just under the breastbone.
L1	Lung: between the ribs, just below the collarbone. Sliding is normal.
A1 / A2	Belly: right and left flank, for liver, kidney and spleen.
A3	Pelvis: just above the pubic bone, for the bladder.

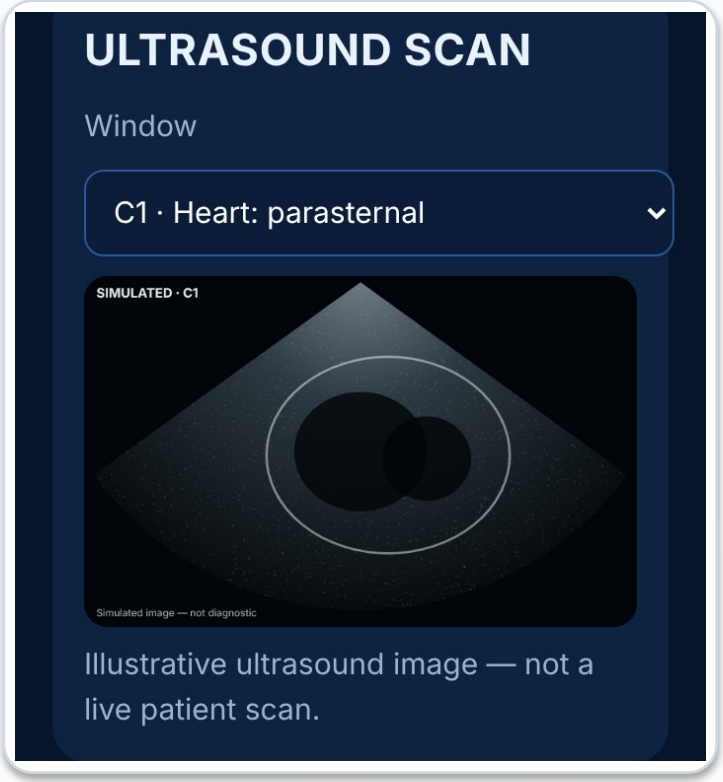
How a crew member places a pad

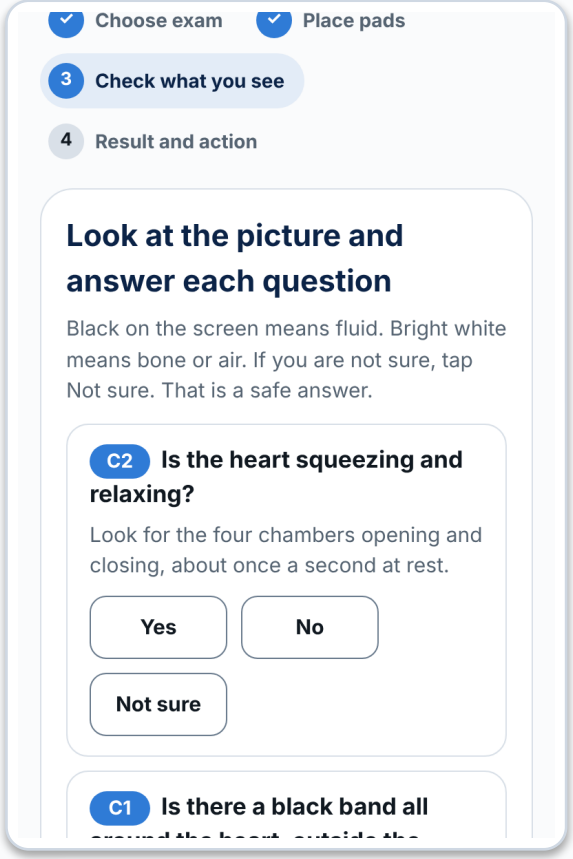
1 Measure from the notch at the base of the throat. 2 Mark the spot. 3 Stick the pad and turn the arrow toward the head. 4 Dock the probe collar. 5 The app shows the pad on screen and asks what you see.

Sample positions for an average adult. A trained sonographer should confirm each window before real use.

Ultrasound simulation

Crew can practise on the ground and in flight. The picture and the questions are the same.





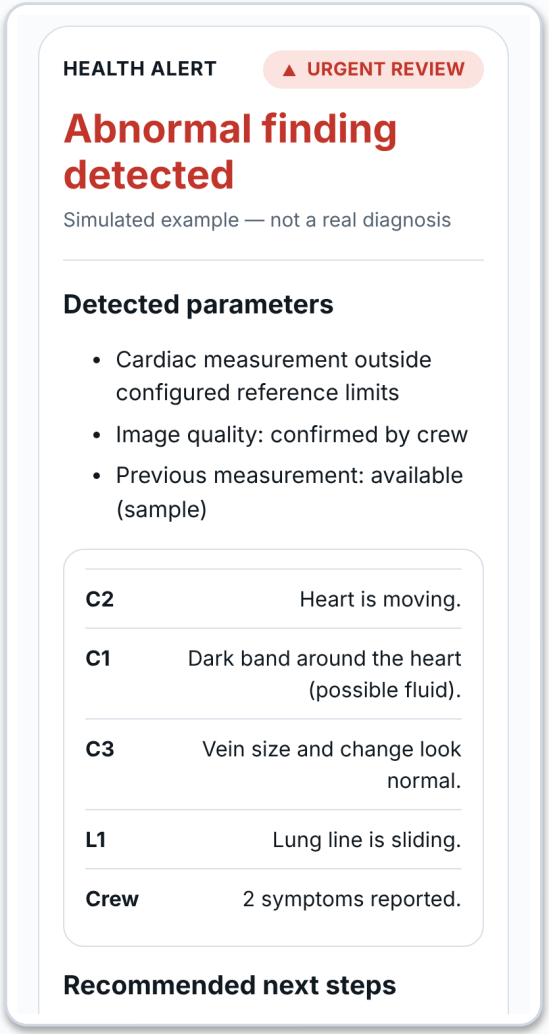
- **Pick a window.** Heart, lung or belly pad.
- **Look at the picture.** Black is fluid. Bright is bone or air.
- **Answer in plain words.** Yes, No or Not sure. Not sure is treated as a concern.
- **Flip a frame.** Capture and add it to the report.

Every image is labelled simulated. The questions follow the same four-view order a trained operator would use: heart motion, fluid around the heart, vein size, lung sliding.

Concept prototype. Simulated data and illustrations. Not a validated medical device.

HEALTH ALERT

The app turns the answers into one clear result: normal, review, or urgent.



NORMAL All four views look normal and no symptoms.

REVIEW One concern, one “not sure”, or a symptom. Send to the flight surgeon.

URGENT No heart movement, two or more findings, or a finding plus a symptom.

What the crew member sees

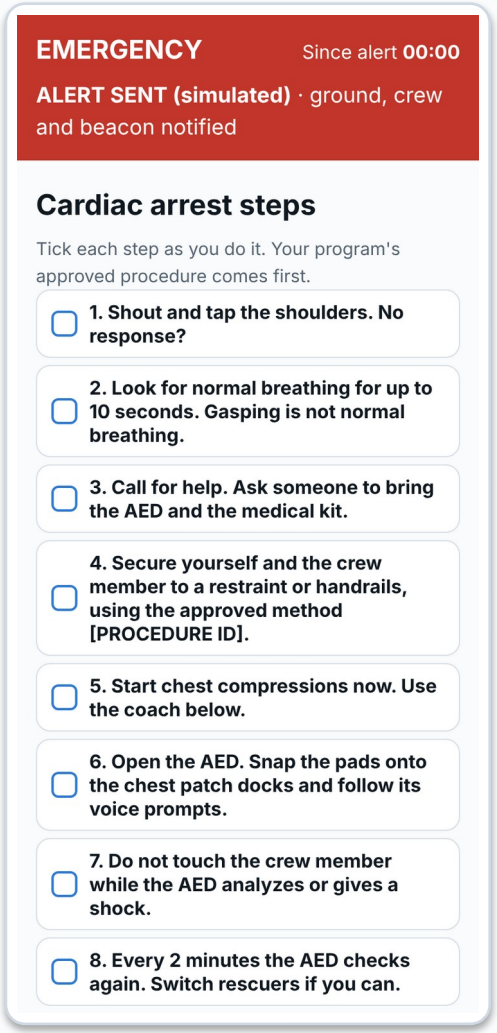
- The result, with each pad and what was seen at it
- Recommended next steps in order
- A button to send the report and alert to the flight surgeon
- The locked medical kit, only if the result is urgent

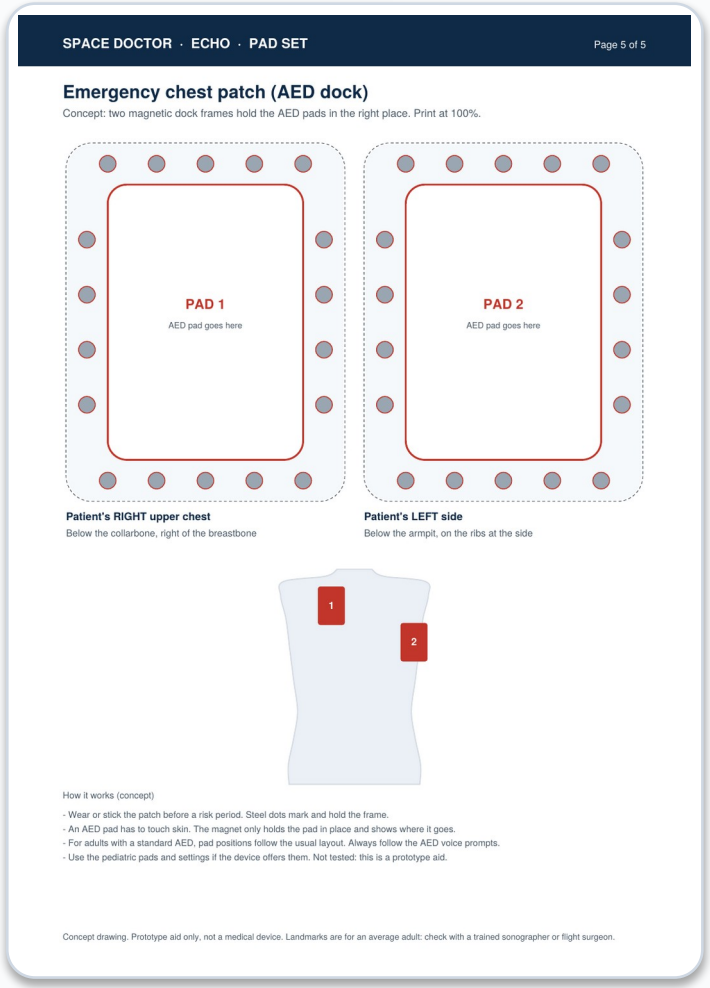
The app never chooses a drug or a dose. The ground team gives the order.

Concept prototype. Simulated data and illustrations. Not a validated medical device.

Guidance for the astronaut

One button for a collapse. Plain steps for someone who has never done it.





- **SOS button.** Always on screen. 10 seconds to cancel, then ground and crew are alerted.
- **CPR coach.** Checklist, then a compression coach: about 100 to 120 per minute, 5 to 6 cm deep, with microgravity methods.
- **AED chest patch.** Two magnetic frames hold the AED pads at the right chest positions. Snap the pads in and follow the voice.
- **Locked kit.** Opens only after a second person authorises it and a five-item check is ticked.
- **Wellness vitals.** Compares heart rate, temperature and blood pressure with the astronaut's own baseline, not an Earth average.

Concept prototype. Simulated data and illustrations. Not a validated medical device.

Thank you

Doctor Space gives a crew member without medical training a way to look, understand, and act.

Impact

- Space: a health backup for crews far from Earth
- Earth: remote clinics and first responders can use the same pad guide
- Free, offline, runs in any browser
- Next: test with a sonographer, then with real crew data

NASA resources used

- 5 Hazards of Human Spaceflight (NASA HRP): distance from Earth, isolation and closed environments shaped the scope
- Risks of Human Spaceflight (NASA HRP): cardiovascular adaptation, sleep and behavior set the vitals tracked
- Ongoing HRP Research in Space

Also: published ISS studies and CSA science pages

Live app: https://alfathitamara.github.io/doctor-space/

Built with help from AI tools (code, text, graphics). Illustrations and sample data are simulated. Not a validated medical device.